

28/10/2015(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
OCT-NOV 2015

Max. Marks: 100

Class: SE

Name of the Course: Applied Mathematics-III

Course Code: UTEC 301

Duration: 3 Hrs

Semester: III

Branch: Computer

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.	Questions	Marks
Q.1	Attempt any THREE	21
(i)	Evaluate: $\int_0^{\infty} e^{-t} \frac{(\cos 3t - \cos 2t)}{t} dt$	
(ii)	Find Inverse Laplace transform of $\frac{s+2}{(s^2+4s+5)^2}$ by using convolution theorem.	
(iii)	Find (a) $L\{(1+3t-4t^2+2t^3)H(t-3)\}$ (b) $L^{-1}\left\{\frac{se^{-as}}{s^2+3s+2}\right\}$	
(iv)	Solve: $y'' + y' - 2y = t$, $y(0) = 1$, $y'(0) = 0$ by L.T method	
(v)	Find Inverse Laplace transform of $\tan^{-1} \frac{1}{s^2}$	
Q.2	Attempt any THREE	21
(i)	Find the Fourier series for $f(x) = x^2$ in $(0, 2\pi)$ and hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$	
(ii)	Find the Fourier series for $f(x) = \cos x $ in $(-\pi, \pi)$	
(iii)	Find half range sine series for $F(x) = \frac{2x}{l}$, $0 \leq x \leq \frac{l}{2}$ $= \frac{2}{l}(l-x)$, $\frac{l}{2} \leq x \leq l$	
(iv)	Find Complex form of Fourier series for $F(X) = \cosh x + \sinh x$ in $(-\pi, \pi)$.	
(v)	Show that the functions $f_1(x) = 1$, $f_2(x) = x$ are orthogonal on $(-1, 1)$. Determine the constants a and b such that the function, $f_3(x) = -1 + ax + bx^2$ is orthogonal to both f_1 and f_2 on that interval.	

28/10/2015(E)

Q.3	Attempt any THREE	21
(i)	Show that the function $v = \tan^{-1} \frac{y}{x}$ is harmonic and find their corresponding analytic function $f(z) = u + iv$	
(ii)	Verify Laplace equation for $u = (r + \frac{a^2}{r}) \cos \theta$. Also find v and $f(z)$.	
(iii)	Find the bilinear transformation which maps the points $0, 1, \infty$ onto the points $-5, -1, 3$.	
(iv)	Show that the image of the rectangular hyperbola $x^2 - y^2 = 1$ under the transformation w is the lemniscate $\rho^2 = \cos 2\phi$.	
(v)	Find the analytic function $f(z) = u + iv$ in terms of z if $3u + 2v = y^2 - x^2 + 16xy$	
Q.4	Attempt any THREE	21
(i)	Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$ hence find $f(r)$ such that $\nabla^2 f(r) = 0$.	
(ii)	Show that $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is irrotational and hence find its scalar potential ϕ . Hence find the work done in moving a particle in this field of force $\vec{F}$ from the point $(1, -2, 1)$ to the point $(3, 1, 4)$	
(iii)	Verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region enclosed by the curves $y = x^2$ and $x = y^2$	
(iv)	Use the divergence theorem to evaluate $\iiint_S (4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}) \cdot d\vec{S}$ where S is the region bounded by the surfaces $y^2 = 4x, x = 1, z = 0, z = 3$	
(v)	Using Stoke's theorem evaluate $\iint_S \nabla \times \vec{A} \cdot d\vec{S}$ where $\vec{A} = (x^2 + y - 4)\hat{i} + 3xy\hat{j} + (2xz + z^2)\hat{k}$ and S is the surface above the xy -plane of the paraboloid $z = 4 - (x^2 + y^2)$	
Q.5	Attempt any TWO	16
(i)	Find the Z transform and the region of convergence of $f(k) = 4^k$ for $k < 0$; $f(k) = 3^k, k \geq 0$	
(ii)	Find the Z transform of $\{\cos \alpha k\}, k \geq 0$, where α is real & also find $Z\{e^{-\alpha k} \cos \beta k\}$	
(iii)	Find the inverse z-transform of $\frac{1}{(z-a)^2}$ if (i) $ z < a$ (ii) $ z > a$	